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Introduction to Databases — Autumn 2025

Exercise 3

(Theory) Hand-in: 05.11.2025 (ADAM, 23:59)
(Practical) Hand-in: 05.11.2025 (during exercise)

Solving the Exercises: The exercises can be solved in small groups of a maximum of two people. Use the notations introduced in the lecture. The DMI plagiarism guidelines apply for this lecture.

Submission Information: Please upload all (Theory) deliverables BEFORE the deadline to ADAM as **a single PDF**. For (Practical) exercise check the **Hand-In** instructions. The solutions must contain your names. Solutions that are handed in too late cannot be considered.

Task (Practical) 1: Visualizing the schema (2 Points)

Visualize the schema of the chess database and derive the *relational schema* from it.

Hand-In: Show the relational schema to one of the assistants/tutors.

Task (Practical) 2: SQLite: Local databases (3 Points)

SQLite is a popular embedded database. In contrast to PostgreSQL which runs in a client-server model (with the client and server usually on different machines), SQLite runs inside the application process and the database is stored locally on disk. The Python standard library already includes bindings to sqlite3 making it easy to use.

We have uploaded a SQLite 3 database dump containing a subset of the data in the PostgreSQL database to SWITCH filesender[1]. Password: `IntroductionToDatabases2025`.

Use the command line utility from the SQLite website[2] to create a database from the dump. You can check that the database was correctly created by opening the database with the command line utility and running the `.sha3sum` command. The result should be `c2efb892126cec41740855dfdd0f856805c43cccb798e00189da5cfa`.

Note: For those familiar with checksums: The `.sha3sum` command calculates a checksum over the logical content of the database, not over the bytes that are actually stored on

[1] `https://filesender.switch.ch/filesender2/?s=download&token=f998b6e7-faaf-41e7-8459-206f079c3c97`, `sha256:c661fd2047b226f31ef5702f620d5327f6cf597757e37fd8c6fb1a672aa79695`
[2] `https://sqlite.org/download.html`

the disk. The layout on disk depends on different factors and is therefore not suitable to determine if two databases are equivalent.

Copy the Python script from last week and replace the PostgreSQL connection with a connection to the SQLite database. Which of the queries still work? Which don't?

Hand-In: Show that you can use the SQLite database from Python and the CLI to the assistants/tutors.

Task (Theory) 3: EER to relational schema (3 Points)

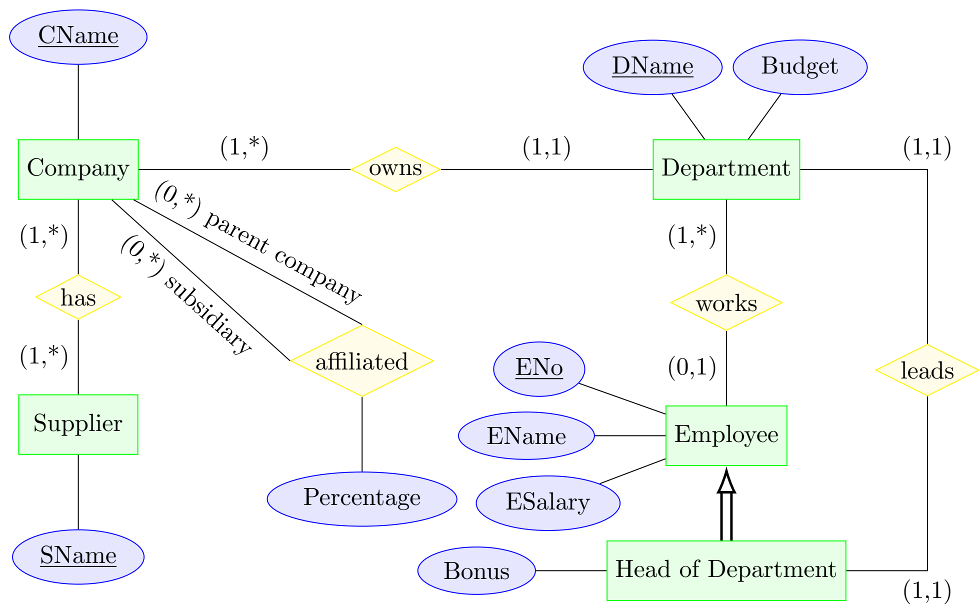

Figure 1: ER diagram

Consider the extended entity-relationship (EER) diagram depicted in Figure 1. Transform the diagram into the relational model and write down the corresponding relational schema. Merge all relations wherever meaningful and possible. Denote all primary and foreign keys.

Hand-In: Upload your relational schema to ADAM.

Task (Theory) 4: Extend an EER diagram (2 Points)

Extend the EER diagram and the relational schema by the fact that an employee is sitting in an office which is located in a specific building. Model this circumstance as precisely as possible.

Hand-In: Upload your adjustements to the EER and the relational schema to ADAM.